Shared quadratic bounds for Diophantine refutation certificates

An explicit quartic in 260 unrestricted integers for Con(ZFC)

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Abstract

We construct and supply a fixed polynomial $P \in \mathbb{Z}[z_1, \ldots, z_{260}]$ of degree four such that ZFC proves that P has no integer zero if and only if ZFC is consistent. The construction develops the repository's direct proof-certificate route; it uses neither a Turing-machine simulation nor a universal-polynomial coefficient tuple. Its main new step replaces eleven individual remainder interval certificates by three shared quadratic bounds. The already available radix margins make each canonical low binomial remainder very small compared with its modulus. A symmetric bound then identifies its signed modular lift, recovering its sign as a consequence. Two adjacent powers of the radix share a bound using one retained square wire and one new sum-of-squares wire.

The principal equation is the sum of 186 completely listed quadratic residual squares. After explicit integral changes of coordinates, it has 4,770 nonzero monomials and coefficient height 9,191. The previous direct endpoint in the inspected incoming archive has 282 variables, 192 residuals, 5,491 monomials and height 128,446. A 261-variable, 185-residual companion is also supplied. We prove the unbounded arithmetic reduction, provide exact reconstruction certificates and independent finite checks, and identify the retained logical calibration. The result is a concrete improvement within this direct construction, not a proof of a global simplicity minimum or of Con(ZFC) itself. No new complete Lean or Rocq proof of the final endpoint is claimed.

Contents

1 The delivered equation and the comparison being made **2**

1.1 A closed integer polynomial . . . 2

1.2 What "simpler" measures here . . . 2

2 A signed remainder can be certified collectively **3**

2.1 One standard nonnegativity representation . . . 3

2.2 The collective bound . . . 3

3 The inherited arithmetic interface **4**

3.1 The proof-certificate route . . . 4

3.2 Semantic coordinates used in the reduction . . . 4

3.3 The three early sign certificates are retained . . . 5

4 Soundness without a circular radix assumption **5**

4.1 Recover powers and arithmetic signs first . . . 5

4.2 A factorial coefficient bound with no factorial premise . . . 6

4.3 The square-radius factorial fixed point . . . 6

4.4 All remaining canonical residues have spare radix capacity . . . 7

5 Completeness and the adjacent-modulus merger **7**

6 Finite algebra: projection, translations and integral shears **8**

7 Why the equivalence is provable in ZFC **8**

7.1 The retained logical chain 8

7.2 A possible well-foundedness pitfall and its resolution 9

7.3 Proof status 9

8 Reproducible artifacts and checks **9**

9 Further research questions **10**

Attribution and novelty scope **10**

A The frozen sequence of integral shears **11**

B Every quadratic residual of the principal equation **12**

1 The delivered equation and the comparison being made

1.1 A closed integer polynomial

The principal equation is

$$\boxed{P(z_1,\dots,z_{260}) = \sum_{j=1}^{186} R_j(z_1,\dots,z_{260})^2 = 0, \qquad (z_1,\dots,z_{260}) \in \mathbb{Z}^{260}.} \tag{1}$$

Every R_j is printed in Appendix B. The sum has a fixed finite length. All coefficients are literal integers; there is no program index, uncomputed numerical port, variable exponent, hidden inequality or additional existential quantifier in (1). A shared polynomial expression in an explanation does not introduce a new unknown. The definitive machine-readable forms are `equations/principal/equation.json` and `equations/principal/expanded_polynomial.json`.

Theorem 1.1 (Consistency equivalence)**.** *For the fixed polynomial in* (1)*, the ordinary mathematical proof given here, composed with the retained ProveIt calibration, is formalizable in ZFC and establishes*

$$\mathrm{ZFC} \;\vdash\; \left[\forall \mathbf{z} \in \mathbb{Z}^{260}\ P(\mathbf{z}) \neq 0\right] \;\longleftrightarrow\; \mathrm{Con(ZFC)}.$$

Here Con(ZFC) *is the usual syntactic assertion that the fixed effective first-order ZFC calculus has no finite proof of contradiction. The same assertion holds for the supplied 261-variable companion.*

The logical calibration is an imported mathematical result, with its interface and internalization discussed in Section 7. The new proof is an existential equivalence of integer zero sets through the arithmetic source. It does not assert that all private witnesses are unique. In particular, a sum-of-three-terms representation of a nonnegative integer usually has several possible witnesses.

1.2 What “simpler” measures here

There is no single agreed ordering of variable count, degree, coefficient height, support and evaluator length. We therefore give the actual measures rather than identify a preferred measure with a global minimum. Coefficient height means the largest absolute value of a *collected* coefficient; support counts nonzero collected monomials. The number of quadratic residuals is a further representation measure, not an arithmetic-operation count.

Equation	Variables	Rows	Degree	Monomials	Height
Inspected direct predecessor	282	192	4	5,491	128,446
Four shared bounds	261	185	4	5,332	128,446
Three shared bounds	260	186	4	5,325	128,446
Principal, after integral shears	260	186	4	4,770	9,191
Wider projection, dimension tradeoff	259	185	4	23,960	240,653

The predecessor is the `strict_dimension_centered` endpoint in the incoming Round 17 archive at commit `e0c98c629bf10d30b695884f6b1d2ab361bdf513` [1, 2]. The repository separately contains equations with far fewer variables, including a 37-integer quartic built through a large fixed coefficient tuple, and high-degree ten-integer forms. We do not claim to improve their variable counts. The present advance is in the direct construction with small, fully materialized coefficients and no universal-polynomial tuple. Likewise, the 261-variable companion keeps a smaller number of residuals than the 260-variable version. The 259-variable alternative saves one further variable but substantially increases support and height; it is supplied as a distinct tradeoff. These distinctions matter when comparing prospective formalization effort and polynomial size.

2 A signed remainder can be certified collectively

2.1 One standard nonnegativity representation

Put

$$T(a,b,c) = a^2 + b^2 + c(c+1).$$

For integers a, b, c, this value is nonnegative. Conversely, every nonnegative integer is represented. Indeed, the three-square theorem represents $4n+1$ by three squares, since this number is never of the excluded form $4^e(8h+7)$. Exactly one square is odd modulo four. Writing the three entries as $2a, 2b, 2c+1$ gives $n = T(a,b,c)$. This familiar conversion is explicitly recorded, for example, in Sun [4, eq. (1.1)]; it is also the conversion used by the retained integer source.

2.2 The collective bound

Lemma 2.1 (Short signed lifts)**.** *Let $L \geq 1$ and $1 \leq m \leq 32$ be integers. Suppose $0 \leq c_i < L/8$ and $r_i \equiv c_i \pmod{L}$. Then*

$$L^2 - 2\sum_{i=1}^{m} r_i^2 - 1 = T(a,b,c) \tag{2}$$

has integer witnesses a, b, c if and only if $r_i = c_i$ for every i.

Proof. If (2) holds, nonnegativity of T gives $2r_i^2 < L^2$, hence $|r_i| < 3L/4$. Thus $|r_i - c_i| < 7L/8 < L$. A multiple of L in this interval must be zero. For the converse, if all $r_i = c_i$, then

$$2\sum_i c_i^2 < \frac{m}{32}L^2 \leq L^2.$$

The integral difference on the left of (2) is therefore nonnegative, and the representation by T supplies the three witnesses. □

This lemma places no prior sign restriction on the proposed remainders. Their signs are recovered from the modular meaning of the equations and the smallness of the canonical residues. The coefficient two is not a cosmetic choice. For $L = 9$, $c = 1$ and $r = -8$, the wrong lift satisfies $r \equiv c \pmod{L}$ and $c < L/8$, while $L^2 - r^2 - 1 = 16 = T(4,0,0)$. Simply dropping the two would invalidate the soundness assertion.

Lemma 2.2 (A quantitative low-digit margin)**.** *Let $q \geq 2$, $k \geq 1$, $n \geq 0$ be integers and suppose $\binom{n}{j} \leq q/16$ for $0 \leq j < k$. With the convention $\binom{n}{j} = 0$ when $j > n$, set*

$$c(n,k) = \sum_{j=0}^{k-1} \binom{n}{j} q^j.$$

Then $0 \le c(n,k) < q^k/8$ *and* $c(n,k) \equiv (q+1)^n \pmod{q^k}$.

Proof. The congruence is the binomial theorem. For the bound,

$$c(n,k) \le \frac{q}{16}\frac{q^k-1}{q-1} < \frac{q^{k+1}}{16(q-1)} \le \frac{q^k}{8}.$$

This argument concerns the canonical low residue and does not presume that a supplied signed remainder already has its intended value. □

Corollary 2.3 (Different moduli with a common scale)**.** *Suppose* $L = s_i L_i$ *with positive integral* s_i, L_i*, and* $r_i \equiv c_i \pmod{L_i}$ *with* $0 \le c_i < L_i/8$*. For at most 32 entries,*

$$L^2 - 2\sum_i (s_i r_i)^2 - 1 = T(a,b,c)$$

has integer witnesses exactly when all $r_i = c_i$*.*

Proof. Apply Lemma 2.1 to $s_i r_i$ and $s_i c_i$ modulo L, then cancel the positive nonzero integers s_i. □

The concrete groups below have sizes $2,5,2,2$ or $7,2,2$, well inside the lemma's limit. There is no assertion that every old interval tuple satisfies the new bound without first using its genuine digit semantics.

3 The inherited arithmetic interface

3.1 The proof-certificate route

The direct source encodes a finite numerical proof calculus using bounded local rows, factorial weights and Chinese-remainder decoding. Its powers are represented by an explicit simultaneous Pell system. Intermediate multiplication outputs are kept as variables so that every residual has degree at most two. Sum-of-squares merging then produces a quartic. The semantic source is a numerical presentation calibrated to C222 and then to ZFC refutability. It is not necessary to run any machine enumerating that calculus.

The source closure in this archive retains the exact constructor files and ordinary source proofs. Several historical presentations use different numeric constructor conventions. The finite source being transformed is the literal signed source stored in `certificate/parent/strict_dimension`; its constructor and calibration files fix that convention. We do not silently interchange the original Cantor codes with the later square-shell codes.

In particular the actual square-shell constructors are

$$I(a,b) = 2((a+b)^2 + a), \quad M(a,b) = I(a,b) + 1,$$

$$A(a,b) = 2((a+b)^2 + 3a + 4b + 3), \quad E(a,b) = A(a,b) + 1.$$

The retained `SHELL_ENCODING_PROOF.md` gives a recursive constructor-preserving bijection to the original Cantor presentation, fixing truth and falsity (0 and 1). It transports derivations in both directions. A guarded disequality witness is reconstructed after recoding; one does not simply apply the code bijection to that arithmetic witness.

3.2 Semantic coordinates used in the reduction

The following letters denote existing semantic values, rather than new quantified coordinates in (1):

$$t = 16 + h_0^2, \qquad V = (F^2+1)^t, \qquad y = V(V-1), \tag{3}$$

$$H_0 = 1 + \sum_{i=1}^{9} A_i, \qquad K = x, \qquad N = B^x, \qquad q = (BH_0)^x. \tag{4}$$

The nine independent CRT representatives are denoted $a, b, c, d, S, S', u, v, C_0$. A fixed positive-constructor expression $h = h(a,b,c,d)$ bounds the rule labels. Every representative and h itself occurs as an exponent in a genuine request with base $q+1$.

The eleven digit rows are listed below. The first is an exact binomial request; the others encode the relevant binomial divisibilities. Their retained numerator identities always imply that the supplied remainder is congruent to $(q+1)^n$ modulo the displayed denominator, even when that remainder and its quotient have not yet been oriented.

Source remainder	Upper index n	Position k	Modulus
`choose0.remainder`	F^2+t	t	q^t
`range8.remainder`	C_0	t	q^t
`range0.remainder`	a	$t+1$	q^{t+1}
`range1.remainder`	b	$t+1$	q^{t+1}
`range2.remainder`	c	$t+1$	q^{t+1}
`range3.remainder`	d	$t+1$	q^{t+1}
`range9.remainder`	h	$t+1$	q^{t+1}
`range4.remainder`	S	$y+1$	q^{y+1}
`range5.remainder`	S'	$y+1$	q^{y+1}
`range6.remainder`	u	x	q^x
`range7.remainder`	v	x	q^x

Previously each remainder r_i had a private slack s_i and an interval certificate

$$L_i = r_i + s_i + 1, \qquad r_i s_i = T(\alpha_i, \beta_i, \gamma_i). \tag{5}$$

Once $L_i > 0$ is known, these equations force both r_i, s_i to be nonnegative. The transformation removes exactly these eleven bound rows and eleven product-certificate rows. The 33 private ternary witnesses and unused slack coordinates are then removed by exact integral projection.

3.3 The three early sign certificates are retained

Let r, s, o denote the separate factorial remainder, factorial slack and exact-binomial output slack. Their three nonnegativity certificates stay in the source, along with

$$B = r + s + 1, \qquad q = B + o + 1, \qquad FB = V + s. \tag{6}$$

The endpoint rows also stay:

$$x = F + 1 + (y+1)u_0, \qquad x = 1 + [1 + (t+1)y]u_t. \tag{7}$$

They are crucial to the argument. Removing the delayed interval certificates does not authorize removing these early certificates.

The inherited orientation-first theorem proves the signed Pell frame and every canonical principal pair before it uses any of (5). The literal deletion audit checks that the interval slacks and ternary witnesses are private: none is an atom needed for the frame, anchor, power requests, or early sign proof. This dependency order is why the transformation is valid.

4 Soundness without a circular radix assumption

4.1 Recover powers and arithmetic signs first

We use the retained orientation-first Pell theorem in its stated signed form. It permits either sign of the large Pell center. It recovers the canonical signed Pell polynomials at the requested indices. If a request has positive base $n \geq 2$ and a negative index $-j$, the power congruence would give $mn^j \equiv 1$ modulo a modulus larger than $|mn^j - 1|$. Consequently $mn^j = 1$, impossible for integer m and $j \geq 1$. The same large-modulus argument gives the exact usual power for nonnegative indices. A request at base one is handled directly when its index is already nonnegative. These are inherited proved properties of the unchanged private system, not new positivity assumptions on its inputs.

From (6), initially $B \geq 1$ and $q \geq 2$, while $t \geq 16$. The shared t pair therefore gives q^t and $V = (F^2+1)^t \geq 1$. The ceiling row gives $F \geq 1$ and

$$FB \geq V \geq F^2 + 1, \qquad B > F, \qquad B \geq 2, \qquad q \geq 3.$$

Thus $V \leq FB < B^2$. The retained cofinal equation gives $y = V(V-1) > F+2$.

The q^x request gives $x \geq 0$. If $u_0 \leq -1$, the first endpoint row would instead give $x \leq F - y < 0$. Hence $u_0 \geq 0$ and $x \geq F+1 \geq 2$. The second endpoint row then forces $u_t \geq 1$, and therefore

$$x \geq 2 + (t+1)y > 6. \tag{8}$$

Every actual CRT representative, and the derived label h, is an exponent at base $q+1 \geq 4$. They are all nonnegative. Hence $H_0 \geq 1$, the base $BH_0 \geq 2$ is valid, and the retained requests give exactly $N = B^x$ and $q = (BH_0)^x$. All remaining requests have their intended arithmetic meanings. The adjacent denominator is obtained from the retained one-step Pell addition and equals $q^{t+1} = q\,q^t$.

In particular, *before* identifying a factorial or a supplied low remainder, we have the stronger radix margin

$$q = (BH_0)^x \geq B^6 \geq 16B^2 > 16V. \tag{9}$$

4.2 A factorial coefficient bound with no factorial premise

Lemma 4.1. *For integers $F, t \geq 1$ and $0 \leq j \leq t$,*

$$\binom{F^2+t}{j} \leq (F^2+1)^t.$$

Proof. If $F^2 \geq t$, binomial coefficients are increasing up to position t, and

$$\binom{F^2+t}{j} \leq \binom{F^2+t}{t} = \prod_{i=1}^{t} \left(1 + \frac{F^2}{i}\right) \leq (F^2+1)^t.$$

If $4 \leq F^2 < t$, each coefficient is at most $2^{F^2+t} < 4^t < (F^2+1)^t$. The remaining case is $F = 1$. Pascal's identity bounds each coefficient of row $t+1$ by the sum of row t, namely 2^t; boundary coefficients satisfy the same inequality. □

Combining this lemma with (9) and Lemma 2.2 shows that the canonical low remainder for the factorial digit is less than $q^t/8$. The shared bound gives its supplied remainder absolute value less than $3q^t/4$. In the merged group this follows first for qr_0 at modulus q^{t+1}, and then by cancellation of the positive q. The unchanged numerator row gives the congruence, so the supplied remainder equals the canonical one.

Only this one component of the bound is needed now. The other ten remainder squares are nonnegative, irrespective of their values. We have not assumed the bounds for their canonical residues in order to decode the factorial digit.

Since $0 \leq B < q$ is independently retained and the coefficient through position t is below q, the exact-binomial numerator identity now gives

$$B = \binom{F^2+t}{t}. \tag{10}$$

4.3 The square-radius factorial fixed point

For completeness of the arithmetic argument we record a short proof of the inherited fixed-point fact used at this stage.

Lemma 4.2. *Suppose $t \geq 16$, $F \geq 1$, $V = (F^2+1)^t$, $B = \binom{F^2+t}{t}$, and nonnegative integers r, s satisfy $B = r+s+1$ and $FB = V + s$. Then $F = t!$. Conversely $F = t!$ supplies such r, s.*

Proof. The two rows are equivalent to $F - 1 < V/B \le F$. Set $G = t!$. Then

$$\frac{V}{B} = G \prod_{i=1}^{t} \frac{F^2 + 1}{F^2 + i} \le G,$$

so $F \le G$. Each factor is nondecreasing in F^2. At $F = 1$ the ratio is $2^t/(t+1)$; hence $F \ge 2^t/(t+1) > t^2$. The last inequality holds at $t = 16$ and persists because $(t+1)(t+2)/t^2 < 2$ for $t \ge 16$.

Using $\prod_i (1 - z_i) \ge 1 - \sum_i z_i$ for $0 \le z_i \le 1$ gives

$$\frac{V}{B} \ge G \left(1 - \sum_{i=1}^{t} \frac{i-1}{F^2 + i}\right) > G \left(1 - \frac{1}{2F}\right),$$

because the sum is at most $t(t-1)/(2F^2) < 1/(2F)$. Since $V/B \le F$,

$$G < \frac{2F^2}{2F - 1} = F + \frac{F}{2F - 1} \le F + 1.$$

Integrality gives $G \le F$, hence $F = G$. Conversely, $G = t! \ge 2^{t-1} > t^2$ for $t \ge 16$ (the last inequality holds at 16 and is preserved by induction). Thus at $F = G$ the same product estimate gives $G - \frac{1}{2} < V/B \le G$. Thus $\lceil V/B \rceil = G$; take $s = GB - V$ and $r = V + B - GB - 1$. Both are nonnegative and satisfy the rows. □

We may now conclude $F = t!$. The reciprocal moduli $d_i = 1 + F^2/i$ are integers, pairwise coprime, and $B = \prod_{i=1}^{t} d_i$. These facts are used only after Lemma 4.2, as in the retained CRT argument.

4.4 All remaining canonical residues have spare radix capacity

The retained index congruence is $B \mid C_0 + F^2 + 1$. Since $t \ge 16$, the product for B is strictly larger than $F^2 + 1$. We already know $C_0 \ge 0$, so $C_0 = 0$ is impossible. Thus $C_0 \ge 1$ and $H_0 \ge 2$.

Every lower position k in the table is positive and at most x by (8). For any ordinary representative $A \ge 0$ and $0 \le j < k$,

$$\binom{A}{j} \le \max(1, A)^j \le H_0^{x-1} \le q/64. \tag{11}$$

The last inequality uses $q = B^x H_0^x$, $B \ge 2$ and $x \ge 6$. The expression $\max(1, A)$ deliberately handles zero representatives.

For the derived-label expression the retained exact constructor majorant is $h \le H_0^{\ell}$ with $\ell = 4,194,304$. This bound also handles zero representatives: if $n = \max(a, b, c, d)$, the positive-constructor estimate uses $\max(1, n) + 1 \le H_0$, because the separate representative C_0 is at least one. The inherited factorial estimate for $t \ge 16$ is $\ell(t+1) < 2F$, while (8) gives $2F < x$. Hence for $j < t + 1$,

$$\binom{h}{j} \le H_0^{\ell t} \le H_0^{x-1} \le q/64. \tag{12}$$

All remaining coefficients needed for low-digit extraction are therefore below $q/16$. Lemma 2.2 gives $c_i < L_i/8$ in every case.

Apply Lemma 2.1, or its scaled version, to each supplied remainder. We obtain $r_i = c_i$ and therefore $0 \le r_i < L_i$ for all eleven entries. Set $s_i = L_i - r_i - 1 \ge 0$. The products $r_i s_i$ are nonnegative, so the standard ternary representation supplies every deleted product certificate in (5). All retained source rows are already unchanged. Thus a new zero extends to a zero of the old signed source, preserving its common coordinates. Its proved CRT and refutation decoding now applies.

5 Completeness and the adjacent-modulus merger

Start with a zero of the old signed source, or obtain one from a finite refutation by its retained constructive completeness proof. The preceding arithmetic facts hold for that zero. In particular all eleven remainders are canonical and $0 \le r_i < L_i/8$.

Grouping equal denominators gives four groups of sizes $2, 5, 2, 2$. Lemma 2.1 provides one ternary certificate for each group. This yields the 261-variable companion. Every old coordinate still needed by another row is kept; only private witnesses and subsequently unused definitions are discarded.

For the principal 260-variable source, write $L_0 = q^t$ and $L = q^{t+1} = qL_0$. Denote the two small-modulus remainders by r_0, r_8 and the five large-modulus remainders by r_a, r_b, r_c, r_d, r_h. The merged seven entries are

$$qr_0,\ qr_8,\ r_a,\ r_b,\ r_c,\ r_d,\ r_h.$$

Each canonical value is less than $L/8$. Keep the existing square wire $Q_2 = q^2$ and define one new wire $S_0 = r_0^2 + r_8^2$. The merged residual is

$$L^2 - 2Q_2S_0 - 2(r_a^2 + r_b^2 + r_c^2 + r_d^2 + r_h^2) - 1 - T(\alpha, \beta, \gamma) = 0. \tag{13}$$

Both this equation and the wire definitions have degree at most two in their stored coordinates. The left side before subtracting T is nonnegative by Lemma 2.1 with seven entries. Choose the three ternary witnesses and retain every other old coordinate. This constructs a new zero. The same lemma proves soundness for the merged bound, including the early factorial stage where the other residue values are still unknown.

Relative to the four-group version, merging removes three bound witnesses and retains two additional wires, saving one integer variable. It leaves one extra residual. The retained q^2 wire had previously been eliminated by projection; it is not a free numerical parameter. The exact variable and row counts are measured after the complete projection, rather than inferred solely from this informal accounting.

6 Finite algebra: projection, translations and integral shears

The quadratic compiler factors the inherited arithmetic rows, substitutes only definitions monic in the eliminated integer coordinate, and records their restoration sequence. A monic definition $x_j = A(\widehat{\mathbf{x}})$ permits integer elimination without a divisibility premise. The checker verifies the full substitutions and that every resulting row is still quadratic. Integral translations are bijections of $\mathbb{Z}^n$.

The secondary coefficient reduction uses only elementary integral shears. For distinct coordinates i, j and $k \in \mathbb{Z}$, replace

$$x_j = z_j + kz_i, \qquad x_v = z_v \quad (v \neq j). \tag{14}$$

This linear map has determinant one and inverse $z_j = x_j - kx_i$. It is a bijection of the integer lattice. Substituting it into every quadratic residual preserves its degree bound and preserves the existence of a common integer zero in both directions. Successive shears compose; the complete ordered sequence appears in Appendix A and in the machine-readable certificate.

The search that chose this sequence is an optimization heuristic. It is not part of the correctness argument. The independent checker replays the frozen sequence by exact coefficient arithmetic, verifies its inverse, then expands all 186 residual squares and compares every coefficient of the delivered polynomial. No assertion that the chosen coordinate basis is optimal is needed. The source and final variable names must also be kept distinct: a final z_i generally is not the original semantic coordinate having the same array index.

Because a finite sum of real squares vanishes exactly when all summands vanish, the resulting single equation has precisely the intended integer zero-set projection. This completes the arithmetic part of Theorem 1.1.

7 Why the equivalence is provable in ZFC

7.1 The retained logical chain

The source chain is the calibrated numerical proof presentation, its C222 interface, ZF refutability, and ZFC refutability. The retained ProveIt declarations include

$$\texttt{KEndpoint.c222_iff_zf_refutation}: \quad C_{222}(1) \longleftrightarrow \mathrm{Ref(ZF)},$$
$$\texttt{ConstructibleChoice.zf_refutation_iff_zfc_refutation}: \quad \mathrm{Ref(ZF)} \longleftrightarrow \mathrm{Ref(ZFC)}.$$

The concrete syntax-convention bridge and arithmetic source construction are preserved in the proof closure. The new transformations neither change the calculus nor assume that one of its derivations exists.

The mathematical logical calibration uses first-order completeness, definable quotient and submodel constructions, and the constructible-model proof of equiconsistency of ZF and ZFC. It should not be described as merely a finite arithmetic calculation. The arithmetic reduction in this paper uses natural-number induction, finite binomial and congruence identities, the three-square theorem, and the inherited explicit Pell and CRT results. These are all statements and arguments available in ZFC.

7.2 A possible well-foundedness pitfall and its resolution

The lower calibration mentions a "well-founded part." An externally well-founded or transitive set model would be an unjustified stronger requirement. The source instead defines *internal* well-foundedness by quantifying over sets represented in the model: every represented nonempty subset of a represented set has an internally minimal element. The files `WellFoundedPartCore.lean`, `WFRelativization.lean` and `WFModelCore.lean` give actual first-order formulas and recursive formula translations for this domain. Schema instances are handled through those translations.

Consequently, inside any model of ZFC the calibration uses its own set structures, natural numbers and internally finite syntax. The definable submodel may be externally ill-founded. No argument replaces the model's natural numbers with the external standard ones, and no transitive-model assumption is introduced. The detailed scoped audit is supplied in `review/c222_internalization_audit.md`.

7.3 Proof status

There are three different claims here. First, the new unbounded zero-set equivalence is proved mathematically in this paper. Second, exact finite programs certify the literal sources, algebraic substitutions, residuals and expanded coefficient files. Third, the logical calibration is an identified retained mathematical dependency whose uniform formalizability in ZFC is audited. These together justify the ordinary ZFC-provability statement of Theorem 1.1.

The release does not provide a newly built end-to-end Lean/Rocq theorem for this final polynomial, or an emitted proof term in the encoded object-language ZFC calculus. The imported calibration's full formal dependency closure has not been rebuilt here. These are precise formalization boundaries, rather than premises asserting consistency. The result supplies the equivalence; it supplies neither an integer zero nor a proof that no integer zero exists.

8 Reproducible artifacts and checks

The archive contains the principal and companion coefficient files, the exact immutable parent inputs, source transformation records, factorization and projection certificates, translations, the integral shear sequence, the mathematical proof closure, and the source of the independent checks. The root `README.md` gives the reproduction commands. Mathematical data are stored with sorted sparse coefficient lists and explicit integer-variable indices.

The checks establish complete polynomial identities by integer arithmetic. They do not infer an identity from random evaluations. The principal polynomial is independently reconstructed as the sum of the delivered quadratic residual squares; its exact degree, height, support and variable liveness are recomputed. Deliberate corruption controls exercise changed coefficients, deleted terms, wrong interval groups and altered coordinate maps. Optimized Python execution is also checked, so correctness does not depend on removable `assert` statements.

The actual source replay authenticates 83 inherited files and checks all three selected arithmetic variants. It rebuilds 16 JSON files per variant byte for byte and rejects 29 deliberate corruptions. A separately written checker, importing neither the producer nor its polynomial implementation, checks the 260- and 261-variable source changes, every monic restoration, and every final coefficient. The independent shear checker verifies all 76 substitutions, both products of the forward and inverse integral coordinate matrices, the reverse residual identity, and three corrupted coordinate/coefficient cases. Normal and optimized Python runs pass. The root verifier additionally identifies the independently checked final rows with the principal file and checks that every printed residual is generated from that same complete list.

The fixed repository comparison is the commit above. The local archive retains the actual source bytes and their

hashes. A later repository change cannot silently change which equation this report describes. Reproducing the finite checks is independent of the truth of Con(ZFC); no numerical search for a refutation is used as evidence of consistency.

9 Further research questions

1. **Share additional denominator groups.** The two remaining moduli are q^{y+1} and q^x. Can an already present quotient or power wire scale one group into another at a lower cost than the three witnesses saved? A new power request must be charged in full. A formal common denominator alone is not a size reduction.
2. **Use the actual margins in a generic compiler.** A compiler could record the modulus, a canonical-residue bound and the stage at which that bound becomes available. It could then select collective certificates automatically, while preserving the dependency order used in Section 4. The signed lifting lemma is a natural reusable formalization target.
3. **Jointly optimize coordinates and projection.** The present shears simplify a completed residual system. Choosing an integral basis before monic projection might expose further eliminations or a lower coefficient height. Such a candidate needs its full inverse coordinate map and a new collected coefficient audit.
4. **Reduce the logical source rather than just its arithmetic.** The direct source still carries a substantial syntax-constructor graph. A different proof presentation may help, but it needs a uniform equiconsistency proof and a measured complete arithmetic encoding. Counting source rules or proof-checker instructions alone does not measure the resulting Diophantine polynomial.
5. **Complete the formal endpoint.** Reify the source theorem, short-lift certificate and explicit algebraic transformations in the repository's formal arithmetic compiler, then derive the fixed polynomial equivalence in the sealed object calculus. The finite files delivered here give precise targets for that work; they do not substitute for it.

Attribution and novelty scope

The numerical proof calculus, logical calibration, factorial/CRT source, Pell representation, signed orientation schedule and earlier compiler belong to the retained ProveIt work and incoming submissions. In particular, the incoming *graph digits* construction already pools natural remainders with a one-sided sum bound. The present result uses a symmetric quadratic bound for unrestricted integer remainders and recovers their signs from short-lift uniqueness. Its contribution is the complete staged application, the adjacent-modulus merger, the explicit smaller quartics, and the additional certified integral coordinate simplification. No literature-wide first-priority claim is made for elementary inequalities, ternary nonnegativity, or integral shears. This note and its new code were prepared with AI assistance; inherited notices and licenses are retained.

References

[1] V. Reshetnikov and contributors, *ProveIt*, research repository, snapshot `e0c98c629bf10d30b695884f6b1d2ab361bdf513`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/e0c98c629bf10d30b695884f6b1d2ab361bdf513/Computability/DiophantineConsistency.

[2] ProveIt incoming research package, *Cofinal Anchors and Height One Equations for ZFC Consistency*, Round 17, `ConZFC_Cofinal_Anchors_Height_One_Round17_2026-10-25.zip`, `docs/incoming` at the snapshot in [1].

[3] ProveIt incoming research package, *Graph digits*, `conzfc_graph_digits_2026-10-24.zip`, `docs/incoming` at the snapshot in [1].

[4] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2026. The nonnegative-integer representation is equation (1.1). https://arxiv.org/abs/2609.01594v2.

[5] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics **64** (2021), 281–306; arXiv:1704.03504v7. https://arxiv.org/abs/1704.03504.

A The frozen sequence of integral shears

There are 76 steps. At step s, the listed triple (i, j, k) means

$$x_j^{(s-1)} = x_j^{(s)} + k x_i^{(s)}, \qquad x_v^{(s-1)} = x_v^{(s)} \ (v \neq j).$$

The final coordinates are $x_v^{(76)} = z_v$. Indices in this table are one-based.

Step	i	j	k
1	11	28	-1
2	11	22	-1
3	13	45	-1
4	28	31	-2
5	45	46	-2
6	60	61	-2
7	26	36	-2
8	12	26	-1
9	22	68	-2
10	49	56	-2
11	28	47	-2
12	26	42	-1
13	28	36	-1
14	13	31	-2
15	55	64	-2
16	52	53	-2
17	26	62	-2
18	57	58	-2
19	28	43	-1
20	63	65	-2
21	13	59	-1
22	26	50	-2
23	13	51	-1
24	13	54	-1
25	22	67	-1
26	26	61	-1
27	33	27	-1
28	11	36	-2
29	11	36	-1
30	13	42	-2
31	13	42	-1
32	13	43	-2
33	13	43	-1
34	13	28	-2
35	13	28	-2
36	27	29	-1
37	12	26	1
38	66	68	-1
39	64	63	-1
40	12	42	-2
41	12	42	-1
42	23	24	-2
43	38	39	-2
44	12	28	-1
45	23	24	-2
46	38	39	-2
47	12	22	-2
48	13	28	2
49	13	26	-1
50	12	22	-1
51	12	26	-1
52	23	24	-2
53	38	39	-2
54	13	36	-2
55	13	36	-2
56	13	33	2
57	13	33	1
58	13	36	-2
59	13	31	1
60	13	34	1
61	12	26	1
62	13	36	1
63	13	47	2
64	11	38	-1
65	23	24	-2
66	38	39	-2
67	47	46	-1
68	11	39	-1
69	33	34	-1
70	33	29	1
71	22	31	-1
72	11	43	-2
73	13	43	1
74	11	43	-1
75	11	24	-2
76	11	24	-1

B Every quadratic residual of the principal equation

All coordinates $z_1, \dots, z_{260}$ are independent unrestricted integers. The following fixed list, together with (1), is the complete equation. The variables in the JSON files use zero-based indices; the printed subscripts below are one-based. This appendix is generated directly from the final residual file and checked against it.

$$R_{1} = -z_{4}^{2} + z_{5}$$

$$R_{2} = z_{1} - z_{1}\,z_{3} - z_{3} + z_{10}$$

$$R_{3} = -1 + z_{2} - z_{3}\,z_{8} - z_{4}$$

$$R_{4} = -1 + z_{2} - z_{9}\,z_{10}$$

$$R_{5} = 1 + z_{5} - z_{6}\,z_{21} + z_{19}$$

$$\begin{aligned} R_{6} = {} & -1 + z_{11} - z_{11}^{2} - 2\,z_{11}\,z_{12} - z_{12} \\ & - z_{12}^{2} + z_{22} \end{aligned}$$

$$\begin{aligned} R_{7} = {} & -5 - 20\,z_{11} - 16\,z_{11}^{2} - 16\,z_{11}\,z_{12} - 16\,z_{11}\,z_{22} \\ & - 10\,z_{12} - 4\,z_{12}^{2} - 8\,z_{12}\,z_{22} - 10\,z_{22} - 4\,z_{22}^{2} \\ & + z_{23} \end{aligned}$$

$$\begin{aligned} R_{8} = {} & -2 - z_{11} - z_{11}^{2} - 4\,z_{11}\,z_{23} - 4\,z_{23} \\ & - 4\,z_{23}^{2} + z_{24} \end{aligned}$$

$$R_{9} = 1 + 2\,z_{24} - 4\,z_{24}^{2} + z_{25}$$

$$\begin{aligned} R_{10} = {} & -2 - z_{12}^{2} - 2\,z_{12}\,z_{13} - z_{13} - z_{13}^{2} \\ & + z_{26} \end{aligned}$$

$$\begin{aligned} R_{11} = {} & -9 + 7\,z_{11} - z_{11}^{2} - 4\,z_{11}\,z_{12} + 4\,z_{11}\,z_{13} \\ & - 4\,z_{11}\,z_{26} + 12\,z_{12} - 4\,z_{12}^{2} + 8\,z_{12}\,z_{13} - 8\,z_{12}\,z_{26} \\ & - 15\,z_{13} - 4\,z_{13}^{2} + 8\,z_{13}\,z_{26} + 12\,z_{26} - 4\,z_{26}^{2} \\ & + z_{27} - z_{33} \end{aligned}$$

$$\begin{aligned} R_{12} = {} & -1 + z_{11} - z_{11}^{2} - 2\,z_{11}\,z_{13} - z_{12} \\ & - z_{13}^{2} + z_{28} \end{aligned}$$

$$\begin{aligned} R_{13} = {} & -4\,z_{11} - 64\,z_{11}^{2} - 64\,z_{11}\,z_{13} - 32\,z_{11}\,z_{22} - 32\,z_{11}\,z_{28} \\ & - 2\,z_{12} - 16\,z_{13}^{2} - 16\,z_{13}\,z_{22} - 16\,z_{13}\,z_{28} - 2\,z_{22} \\ & - 4\,z_{22}^{2} - 8\,z_{22}\,z_{28} - z_{27} - 4\,z_{28}^{2} + z_{29} \\ & + z_{33} \end{aligned}$$

$$R_{14} = -36\,z_{13}^{2} + 24\,z_{13}\,z_{29} - 4\,z_{29}^{2} + z_{30}$$

$$\begin{aligned} R_{15} = {} & -6 - 8\,z_{11} - 4\,z_{11}^{2} - 8\,z_{11}\,z_{12} - 8\,z_{11}\,z_{13} \\ & + 8\,z_{11}\,z_{26} + 8\,z_{11}\,z_{28} - 6\,z_{12} - 4\,z_{12}^{2} - 8\,z_{12}\,z_{13} \\ & + 8\,z_{12}\,z_{26} + 8\,z_{12}\,z_{28} - 5\,z_{13} - 4\,z_{13}^{2} + 8\,z_{13}\,z_{26} \\ & + 8\,z_{13}\,z_{28} - z_{22} + 8\,z_{26} - 4\,z_{26}^{2} - 8\,z_{26}\,z_{28} \\ & + 6\,z_{28} - 4\,z_{28}^{2} + z_{31} \end{aligned}$$

$$\begin{aligned} R_{16} = {} & 1 - 4\,z_{11} + 2\,z_{12} - 4\,z_{12}^{2} - 8\,z_{12}\,z_{13} \\ & - 8\,z_{12}\,z_{31} + 4\,z_{13} - 4\,z_{13}^{2} - 8\,z_{13}\,z_{31} - 2\,z_{22} \\ & + 4\,z_{31} - 4\,z_{31}^{2} + z_{32} \end{aligned}$$

$$\begin{aligned} R_{17} = {} & -10 - 5\,z_{11} - z_{11}^{2} - 12\,z_{11}\,z_{12} - 12\,z_{11}\,z_{13} \\ & - 4\,z_{11}\,z_{26} - 36\,z_{12} - 36\,z_{12}^{2} - 72\,z_{12}\,z_{13} - 24\,z_{12}\,z_{26} \\ & - 33\,z_{13} - 36\,z_{13}^{2} - 24\,z_{13}\,z_{26} - 12\,z_{26} - 4\,z_{26}^{2} \\ & + z_{33} \end{aligned}$$

$$R_{18} = -2 - 16\,z_{11} - 16\,z_{11}^2 + 8\,z_{11}\,z_{12} - 32\,z_{11}\,z_{13} - 16\,z_{11}\,z_{28} + 5\,z_{12} - z_{12}^2 + 8\,z_{12}\,z_{13} + 4\,z_{12}\,z_{28} - 15\,z_{13} - 16\,z_{13}^2 - 16\,z_{13}\,z_{28} - 8\,z_{28} - 4\,z_{28}^2 - z_{33} + z_{34}$$

$$R_{19} = 2 - 64\,z_{13}^2 - 32\,z_{13}\,z_{34} - 4\,z_{34}^2 + z_{35}$$

$$R_{20} = -12 - 3\,z_{11} - 47\,z_{12} - 36\,z_{12}^2 + 24\,z_{12}\,z_{22} + 24\,z_{12}\,z_{26} - z_{13} + 16\,z_{22} - 4\,z_{22}^2 - 8\,z_{22}\,z_{26} + 14\,z_{26} - 4\,z_{26}^2 - z_{28} + z_{36}$$

$$R_{21} = -3 - 6\,z_{11} + 2\,z_{12} - 2\,z_{13} - 2\,z_{28} - 4\,z_{36} - 4\,z_{36}^2 + z_{37}$$

$$R_{22} = -1 - 13\,z_{11} - 16\,z_{11}^2 - 16\,z_{11}\,z_{12} - 16\,z_{11}\,z_{22} - 6\,z_{12} - 4\,z_{12}^2 - 8\,z_{12}\,z_{22} - 6\,z_{22} - 4\,z_{22}^2 + z_{38}$$

$$R_{23} = -2 + 3\,z_{11} - z_{11}^2 + 4\,z_{11}\,z_{38} - 4\,z_{38} - 4\,z_{38}^2 + z_{39}$$

$$R_{24} = 1 + 2\,z_{39} - 4\,z_{39}^2 + z_{40}$$

$$R_{25} = -z_{12}^2 - 4\,z_{12}\,z_{40} - 4\,z_{40}^2 + z_{41}$$

$$R_{26} = -6 - 16\,z_{11} - 16\,z_{11}^2 + 32\,z_{11}\,z_{12} - 32\,z_{11}\,z_{22} + 6\,z_{12} - 16\,z_{12}^2 + 32\,z_{12}\,z_{22} - 2\,z_{13} - 14\,z_{22} - 16\,z_{22}^2 - z_{26} + z_{42}$$

$$R_{27} = -4 - 3\,z_{11} - 7\,z_{12} - 4\,z_{13} - 2\,z_{26} - z_{28} + 4\,z_{42} - 4\,z_{42}^2 + z_{43}$$

$$R_{28} = -6\,z_{11} + 2\,z_{12} + 2\,z_{13} - 4\,z_{13}^2 - 8\,z_{13}\,z_{43} - 2\,z_{28} + 4\,z_{43} - 4\,z_{43}^2 + z_{44}$$

$$R_{29} = -z_{13} - z_{13}^2 - 2\,z_{13}\,z_{14} - z_{14}^2 + z_{45}$$

$$R_{30} = 2\,z_{11} - z_{11}^2 - 4\,z_{11}\,z_{45} + 2\,z_{45} - 4\,z_{45}^2 + z_{46} - z_{47}$$

$$R_{31} = -35 + 2\,z_{12} - 66\,z_{13} - 36\,z_{13}^2 + 24\,z_{13}\,z_{26} + 24\,z_{13}\,z_{28} + 24\,z_{26} - 4\,z_{26}^2 - 8\,z_{26}\,z_{28} + 22\,z_{28} - 4\,z_{28}^2 + z_{47}$$

$$R_{32} = -16\,z_{13}^2 - 16\,z_{13}\,z_{46} - 2\,z_{46} - 4\,z_{46}^2 + 2\,z_{47} + z_{48}$$

$$R_{33} = -16\,z_{11}^2 + 24\,z_{11}\,z_{13} - 16\,z_{11}\,z_{22} + 16\,z_{11}\,z_{31} + z_{13} - 9\,z_{13}^2 + 12\,z_{13}\,z_{22} - 12\,z_{13}\,z_{31} - 4\,z_{22}^2 + 8\,z_{22}\,z_{31} - 4\,z_{31}^2 + z_{49}$$

$$R_{34} = -1 + 6\,z_{12} - 4\,z_{12}^2 + 8\,z_{12}\,z_{13} - 8\,z_{12}\,z_{26} - 6\,z_{13} - 4\,z_{13}^2 + 8\,z_{13}\,z_{26} + 6\,z_{26} - 4\,z_{26}^2 + z_{50}$$

$$R_{35} = -1 - z_{13} - z_{13}^2 - 4\,z_{13}\,z_{50} - 4\,z_{50}^2 + z_{51}$$

$$R_{36} = 1 + z_{12} - z_{12}^2 - 4\,z_{12}\,z_{51} - 4\,z_{51}^2 + z_{52}$$

$$R_{37} = -1 - 6\,z_{52} - 4\,z_{52}^2 + z_{53}$$

$$R_{38} = -1 - z_{13} - z_{13}^2 - 4\,z_{13}\,z_{53} - 4\,z_{53}^2 + z_{54}$$

$$R_{39} = 2 + z_{12} - z_{12}^2 - 4\,z_{12}\,z_{54} - 4\,z_{54}^2 + z_{55}$$

$$R_{40} = -2\,z_{49} - 4\,z_{49}^2 - 8\,z_{49}\,z_{55} - 4\,z_{55}^2 + z_{56}$$

$$R_{41} = -9 + 7\,z_{11} - z_{11}^2 - 4\,z_{11}\,z_{56} + 12\,z_{56} - 4\,z_{56}^2 + z_{57}$$

$$R_{42} = -1 - 6\,z_{57} - 4\,z_{57}^2 + z_{58}$$

$$R_{43} = -z_{13} - z_{13}^2 - 4\,z_{13}\,z_{58} - 4\,z_{58}^2 + z_{59}$$

$$R_{44} = -z_{11} - z_{11}^2 - 4\,z_{11}\,z_{59} - 4\,z_{59} - 4\,z_{59}^2 + z_{60}$$

$$R_{45} = -10 - z_{12} + z_{13} - z_{26} - 14\,z_{60} - 4\,z_{60}^2 + z_{61}$$

$$R_{46} = 1 - 2\,z_{12} + 2\,z_{13} - 2\,z_{26} + 4\,z_{61} - 4\,z_{61}^2 + z_{62}$$

$$R_{47} = z_{12} - z_{12}^2 - 4\,z_{12}\,z_{62} - 4\,z_{62}^2 + z_{63} - z_{64}$$

$$R_{48} = -10 - 14\,z_{55} - 4\,z_{55}^2 + z_{64}$$

$$R_{49} = 1 - 2\,z_{63} - 4\,z_{63}^2 + 2\,z_{64} + z_{65}$$

$$R_{50} = z_{13} - z_{13}^2 - 4\,z_{13}\,z_{65} - 4\,z_{65}^2 + z_{66}$$

$$R_{51} = 15\,z_{12} - 36\,z_{12}^2 + 12\,z_{12}\,z_{13} + 24\,z_{12}\,z_{22} - z_{13} - z_{13}^2 - 4\,z_{13}\,z_{22} - 5\,z_{22} - 4\,z_{22}^2 + z_{67}$$

$$R_{52} = 1 + 6\,z_{12} - 2\,z_{22} - z_{66} + 4\,z_{67} - 4\,z_{67}^2 + z_{68}$$

$$R_{53} = -2 + 2\,z_{66} - 2\,z_{68} - 4\,z_{68}^2 + z_{69}$$

$$R_{54} = -2 - 4\,z_{69} - 4\,z_{69}^2 + z_{70}$$

$$R_{55} = -z_4\,z_{16} + z_{71}$$

$$R_{56} = z_{15} - z_{16} - 2\,z_{25}\,z_{71} - z_{72}$$

$$R_{57} = 12\,z_{13}\,z_{71} + z_{15} - z_{16} - 4\,z_{27}\,z_{71} - 2\,z_{30}\,z_{71} + 4\,z_{33}\,z_{71} - z_{73}$$

$$R_{58} = z_{15} - z_{16} - 2\,z_{32}\,z_{71} - 6\,z_{71} - z_{74}$$

$$R_{59} = -12\,z_{13}\,z_{71} + z_{15} - z_{16} - 4\,z_{33}\,z_{71} - 2\,z_{35}\,z_{71} - z_{75}$$

$$R_{60} = z_{15} - z_{16} - 2\,z_{37}\,z_{71} + 6\,z_{71} - z_{76}$$

$$R_{61} = z_{15} - z_{16} - 4\,z_{40}\,z_{71} - 2\,z_{41}\,z_{71} - z_{77}$$

$$R_{62} = z_{15} - z_{16} - 2\,z_{44}\,z_{71} - z_{78}$$

$$R_{63} = z_{15} - z_{16} - 2\,z_{48}\,z_{71} - 4\,z_{71} - z_{79}$$

$$R_{64} = -z_{11}\,z_{12} + z_{11}\,z_{13} + z_{12} + z_{12}\,z_{13} - z_{13} - z_{13}^2 + z_{80}$$

$$R_{65} = 1 + 2\,z_{14} + z_{14}^2 - z_{80}^2 + z_{81}$$

$$R_{66} = -z_{81}^2 + z_{82}$$

$$R_{67} = -4\,z_{69}\,z_{71} - 2\,z_{70}\,z_{71} + 2\,z_{71} + z_{83}$$

$$\begin{aligned} R_{68} = {} & -z_{15}^2 + 2\,z_{15}\,z_{16} + 2\,z_{15}\,z_{83} - z_{16}^2 - 2\,z_{16}\,z_{83} \\ & - z_{83}^2 + z_{84} \end{aligned}$$

$$R_{69} = -1 - z_4\,z_{12} + z_{85}$$

$$R_{70} = -z_{16}\,z_{85} + z_{87}\,z_{88} + z_{89}$$

$$R_{71} = -1 + z_4 - z_4\,z_{11} + z_{86}$$

$$R_{72} = -1 + 2\,z_4 + 6\,z_4\,z_{12} - 2\,z_4\,z_{22} + z_{87}$$

$$R_{73} = -z_{15}\,z_{86} + z_{88}$$

$$R_{74} = -1 - 4\,z_4\,z_{11} - 2\,z_4\,z_{12} - 2\,z_4\,z_{22} + z_{90}$$

$$R_{75} = z_{15}\,z_{85} - z_{16}\,z_{90} + z_{91}$$

$$R_{76} = -z_{15}\,z_{72} + z_{16}\,z_{72} + z_{92}$$

$$R_{77} = -z_{73}\,z_{92} + z_{93}$$

$$R_{78} = -z_{74}\,z_{93} + z_{94}$$

$$R_{79} = -z_{75}\,z_{94} + z_{95}$$

$$R_{80} = -z_{76}\,z_{95} + z_{96}$$

$$R_{81} = -z_{77}\,z_{96} + z_{97}$$

$$R_{82} = -z_{78}\,z_{97} + z_{98}$$

$$R_{83} = -z_{79}\,z_{98} + z_{99}$$

$$R_{84} = -z_{89}\,z_{99} + z_{100}$$

$$R_{85} = -z_{91}\,z_{100} + z_{101}$$

$$R_{86} = -z_{101}^2 + z_{102}$$

$$R_{87} = -z_3 - z_3\,z_{19} + z_{19} + z_{103}$$

$$R_{88} = -z_{17}\,z_{103} + z_{104}$$

$$\begin{aligned} R_{89} = {} & -2\,z_2^2 + 2\,z_2\,z_{15} + 2\,z_2\,z_{16} + 2\,z_2\,z_{104} + 2\,z_2\,z_{105} \\ & - z_{15}^2 - 2\,z_{15}\,z_{104} - z_{16}^2 - 2\,z_{16}\,z_{105} + z_{20}\,z_{107} \\ & - z_{82}\,z_{102} - z_{84}\,z_{102} - z_{104}^2 - z_{105}^2 \end{aligned}$$

$$R_{90} = -z_3\,z_{18} + z_{18} - z_{18}\,z_{103} + z_{105}$$

$$\begin{aligned} R_{91} = {} & -5\,z_{11} + 3\,z_{12} - 4\,z_{22} - 2\,z_{25} - 4\,z_{27} \\ & - 2\,z_{30} - 2\,z_{32} - 2\,z_{35} - 2\,z_{37} - 4\,z_{40} \\ & - 2\,z_{41} - 2\,z_{44} - 2\,z_{48} - 4\,z_{69} - 2\,z_{70} \\ & + z_{106} \end{aligned}$$

$$R_{92} = -z_{108}\,z_{111} + z_{111} + z_{113}$$

$$R_{93} = -z_6\,z_{110} + z_{109} - z_{110}\,z_{113} - z_{112}$$

$$R_{94} = -z_{20}\,z_{115} + z_{118}$$

$$R_{95} = z_{114} - z_{116}\,z_{118} - z_{117}$$

$$R_{96} = -z_{118}\,z_{120} + z_{119} - z_{121}$$

$$R_{97} = -z_{118}\,z_{123} + z_{122} - z_{124}$$

$$R_{98} = -z_{118}\,z_{126} + z_{125} - z_{127}$$

$$R_{99} = -z_{20}\, z_{129} + z_{132}$$

$$R_{100} = z_{128} - z_{130}\, z_{132} - z_{131}$$

$$R_{101} = -z_{132}\, z_{134} + z_{133} - z_{135}$$

$$R_{102} = -z_{20}\, z_{137} + z_{140}$$

$$R_{103} = z_{136} - z_{138}\, z_{140} - z_{139}$$

$$R_{104} = -z_{140}\, z_{142} + z_{141} - z_{143}$$

$$R_{105} = -z_{20}\, z_{110} + z_{147}$$

$$R_{106} = z_{144} - z_{145}\, z_{147} - z_{146}$$

$$R_{107} = -z_{118}\, z_{149} + z_{148} - z_{150}$$

$$R_{108} = 7 + z_{151} - 2\, z_{151}\, z_{153} - 14\, z_{153} + z_{154}$$

$$R_{109} = -z_{151}\, z_{154} - 8\, z_{154} + z_{155}$$

$$R_{110} = 2\, z_{152} + z_{152}^2 + z_{154}^2 - z_{155}^2$$

$$R_{111} = -z_{152}\, z_{157} - z_{157} + z_{158}$$

$$R_{112} = 2\, z_{156} + z_{156}^2 + z_{157}^2 - z_{158}^2$$

$$R_{113} = z_5^2 - 2\, z_5\, z_{152} - 2\, z_{152} + z_{159}$$

$$R_{114} = -1 - z_5\, z_{157} + z_7 - z_{156} - z_{157} + z_{158} + z_{159}\, z_{160}$$

$$R_{115} = 1 - 2\, z_6 + z_6^2 - 2\, z_6\, z_{152} + z_{161}$$

$$R_{116} = -z_6\, z_{208} + z_{20} + z_{161}\, z_{162} - z_{207} + z_{209}$$

$$R_{117} = -1 + 2\, z_{152}\, z_{223} - z_{163} + 2\, z_{223} - z_{223}^2$$

$$R_{118} = -1 + z_{108} + z_{163}\, z_{164} - z_{207} - z_{208}\, z_{223} + z_{209}$$

$$R_{119} = -z_{152}\, z_{166} - z_{166} + z_{167}$$

$$R_{120} = 2\, z_{165} + z_{165}^2 + z_{166}^2 - z_{167}^2$$

$$R_{121} = 1 - 2\, z_{108} + z_{108}^2 - 2\, z_{108}\, z_{152} + z_{168}$$

$$R_{122} = -1 - z_{108}\, z_{166} + z_{109} - z_{165} + z_{167} + z_{168}\, z_{169}$$

$$R_{123} = 3 - 2\, z_{108} - 2\, z_{108}\, z_{152} + 2\, z_{152} + z_{170} + z_{171}$$

$$R_{124} = -1 + 2\, z_{108} - z_{108}^2 + z_{171}$$

$$R_{125} = -1 - z_{108}\, z_{157} + z_{110} - z_{156} + z_{157} + z_{158} + z_{170}\, z_{172}$$

$$R_{126} = 1 + z_{152} - z_{152}\, z_{174} - z_{174} + z_{175}$$

$$R_{127} = 1 + 2\, z_{173} + z_{173}^2 - 2\, z_{174} + z_{174}^2 - z_{175}^2$$

$$R_{128} = -1 + z_{108} - z_{108}\, z_{174} + z_{114} + z_{168}\, z_{176} - z_{173} + z_{175}$$

$$R_{129} = -2\, z_{152} - z_{152}^2 + z_{177}$$

$$R_{130} = 1 - z_{108} - z_{108}\, z_{156} - z_{108}\, z_{158} + z_{115} + z_{152}\, z_{158} + z_{156} - z_{157}\, z_{177} + 2\, z_{158} + z_{170}\, z_{178}$$

$$R_{131} = -z_{152}\, z_{180} - z_{180} + z_{181}$$

$$R_{132} = 2\, z_{179} + z_{179}^2 + z_{180}^2 - z_{181}^2$$

$$R_{133} = -1 - z_{108}\, z_{180} + z_{119} + z_{168}\, z_{182} - z_{179} + z_{181}$$

$$R_{134} = -z_{152}\, z_{184} - z_{184} + z_{185}$$

$$R_{135} = 2\, z_{183} + z_{183}^2 + z_{184}^2 - z_{185}^2$$

$$R_{136} = -1 - z_{108}\, z_{184} + z_{122} + z_{168}\, z_{186} - z_{183} + z_{185}$$

$$R_{137} = -z_{152}\, z_{188} - z_{188} + z_{189}$$

$$R_{138} = 2\, z_{187} + z_{187}^2 + z_{188}^2 - z_{189}^2$$

$$R_{139} = -1 - z_{108}\, z_{188} + z_{125} + z_{168}\, z_{190} - z_{187} + z_{189}$$

$$R_{140} = -z_{152}\, z_{192} - z_{192} + z_{193}$$

$$R_{141} = 2\, z_{191} + z_{191}^2 + z_{192}^2 - z_{193}^2$$

$$R_{142} = -1 - z_{108}\, z_{192} + z_{128} + z_{168}\, z_{194} - z_{191} + z_{193}$$

$$R_{143} = -z_{152}\, z_{196} - z_{196} + z_{197}$$

$$R_{144} = 2\, z_{195} + z_{195}^2 + z_{196}^2 - z_{197}^2$$

$$R_{145} = -1 - z_{108}\, z_{196} + z_{129} + z_{170}\, z_{198} - z_{195} + z_{196} + z_{197}$$

$$R_{146} = -z_{152}\, z_{200} - z_{200} + z_{201}$$

$$R_{147} = 2\, z_{199} + z_{199}^2 + z_{200}^2 - z_{201}^2$$

$$R_{148} = -1 - z_{108}\, z_{200} + z_{133} + z_{168}\, z_{202} - z_{199} + z_{201}$$

$$R_{149} = -z_{152}\, z_{204} - z_{204} + z_{205}$$

$$R_{150} = 2\, z_{203} + z_{203}^2 + z_{204}^2 - z_{205}^2$$

$$R_{151} = -1 - z_{108}\, z_{204} + z_{136} + z_{168}\, z_{206} - z_{203} + z_{205}$$

$$R_{152} = 1 - z_{152}\, z_{208} - z_{208} + z_{209}$$

$$R_{153} = -1 + 2\, z_{207} + z_{207}^2 + z_{208}^2 - 2\, z_{209} - z_{209}^2$$

$$R_{154} = -z_{108}\, z_{208} + z_{137} + z_{170}\, z_{210} - z_{207} + z_{208} + z_{209}$$

$$R_{155} = -z_{152}\, z_{212} - z_{212} + z_{213}$$

$$R_{156} = 2\, z_{211} + z_{211}^2 + z_{212}^2 - z_{213}^2$$

$$R_{157} = -1 - z_{108}\, z_{212} + z_{141} + z_{168}\, z_{214} - z_{211} + z_{213}$$

$$R_{158} = -z_{152}\, z_{216} - z_{216} + z_{217}$$

$$R_{159} = 2\, z_{215} + z_{215}^2 + z_{216}^2 - z_{217}^2$$

$$R_{160} = -1 - z_{108}\, z_{216} + z_{144} + z_{168}\, z_{218} - z_{215} + z_{217}$$

$$R_{161} = -z_{152}\, z_{220} - z_{220} + z_{221}$$

$$R_{162} = 2\, z_{219} + z_{219}^2 + z_{220}^2 - z_{221}^2$$

$$R_{163} = -1 - z_{108}\, z_{220} + z_{148} + z_{168}\, z_{222} - z_{219} + z_{221}$$

$$\begin{aligned} R_{164} = {} & 1 - 2\, z_1 - 3\, z_1^2 - 2\, z_1\, z_5 - z_2^2 \\ & - z_3^2 - 2\, z_5 - 2\, z_5^2 - z_6^2 - z_7^2 \\ & + 2\, z_{11} - z_{11}^2 - z_{12}^2 - z_{13}^2 - z_{14}^2 \\ & - z_{15}^2 - z_{16}^2 - z_{17}^2 - z_{18}^2 - z_{19}^2 \\ & - z_{20}^2 - z_{106}^2 + 2\, z_{108} - 2\, z_{108}^2 - z_{109}^2 \\ & - z_{110}^2 - z_{114}^2 - z_{115}^2 - z_{119}^2 - z_{122}^2 \\ & - z_{125}^2 - z_{128}^2 - z_{129}^2 - z_{133}^2 - z_{136}^2 \\ & - z_{137}^2 - z_{141}^2 - z_{144}^2 - z_{148}^2 + z_{151} \\ & - z_{223}^2 \end{aligned}$$

$$R_{165} = -z_6\, z_{11} - z_6\, z_{12} - z_6\, z_{13} - z_6\, z_{14} - z_6\, z_{15} - z_6\, z_{16} - z_6\, z_{17} - z_6\, z_{18} - z_6\, z_{19} + z_{223}$$

$$R_{166} = -16 + z_1 - z_{224}^2$$

$$R_{167} = -1 + z_3 + z_7 - z_7^2$$

$$R_{168} = -z_{152}\, z_{226} - z_{226} + z_{235}$$

$$R_{169} = -1 + z_{225}^2 + z_{226}^2 - z_{235}^2$$

$$R_{170} = -z_{228} - z_{228}\, z_{229} + z_{236}$$

$$R_{171} = -1 + z_{227}^2 + z_{228}^2 - z_{236}^2$$

$$R_{172} = -z_{220}^2 + z_{237}$$

$$R_{173} = z_{226} - 2\, z_{231}\, z_{237} - z_{237}$$

$$R_{174} = -4\, z_{226} - 8\, z_{226}\, z_{230} + z_{229}$$

$$R_{175} = -z_{152} - z_{225}\, z_{232} + z_{229}$$

$$R_{176} = -1 - z_{219} - z_{225}\, z_{233} + z_{227}$$

$$R_{177} = -z_{106} - 4\, z_{220}\, z_{234} + z_{228}$$

$$\begin{aligned} R_{178} = {} & 2\, z_1^2 + 2\, z_1\, z_5 - 2\, z_1\, z_{157} - 2\, z_1\, z_{166} + z_2^2 \\ & - 2\, z_2\, z_{208} + z_3^2 - 2\, z_3\, z_{196} + z_5^2 - 2\, z_5\, z_{166} \\ & + z_{11}^2 - 2\, z_{11}\, z_{174} + z_{12}^2 - 2\, z_{12}\, z_{180} + z_{13}^2 \\ & - 2\, z_{13}\, z_{184} + z_{14}^2 - 2\, z_{14}\, z_{188} + z_{15}^2 - 2\, z_{15}\, z_{192} \\ & + z_{16}^2 - 2\, z_{16}\, z_{200} + z_{17}^2 - 2\, z_{17}\, z_{204} + z_{18}^2 \\ & - 2\, z_{18}\, z_{212} + z_{19}^2 - 2\, z_{19}\, z_{216} + z_{106}^2 - 2\, z_{106}\, z_{220} \\ & - z_{152}\, z_{238} + z_{156}^2 + z_{157}^2 + z_{165}^2 + z_{166}^2 \\ & + z_{173}^2 + z_{174}^2 + z_{179}^2 + z_{180}^2 + z_{183}^2 \\ & + z_{184}^2 + z_{187}^2 + z_{188}^2 + z_{191}^2 + z_{192}^2 \\ & + z_{195}^2 + z_{196}^2 + z_{199}^2 + z_{200}^2 + z_{203}^2 \\ & + z_{204}^2 + z_{207}^2 + z_{208}^2 + z_{211}^2 + z_{212}^2 \\ & + z_{215}^2 + z_{216}^2 + z_{219}^2 + z_{220}^2 \end{aligned}$$

$$R_{179} = -1 - z_4\, z_6 + z_6 + z_7 - z_{239}^2 - z_{240}^2 - z_{241} - z_{241}^2$$

$$R_{180} = z_4\, z_6 - z_7 - z_{242}^2 - z_{243}^2 - z_{244} - z_{244}^2$$

$$R_{181} = -2 - z_6 + z_{108} - z_{245}^2 - z_{246}^2 - z_{247} - z_{247}^2$$

$$\begin{aligned} R_{182} = -2 &- z_{106}^2 - z_{157}^2 - z_{166}^2 + 2\, z_{174} \\ &- z_{174}^2 - z_{180}^2 - z_{184}^2 - z_{188}^2 - z_{192}^2 \\ &- z_{196}^2 - z_{200}^2 - z_{204}^2 - z_{208}^2 - z_{212}^2 \\ &- z_{216}^2 + z_{220}^2 - z_{248}^2 - z_{249}^2 - z_{250} \\ &- z_{250}^2 \end{aligned}$$

$$R_{183} = -z_{112}^2 - z_{146}^2 + z_{251}$$

$$\begin{aligned} R_{184} = -1 &+ z_{115}^2 - 2\, z_{117}^2 - 2\, z_{121}^2 - 2\, z_{124}^2 \\ &- 2\, z_{127}^2 - 2\, z_{150}^2 - 2\, z_{171}\, z_{251} - z_{252}^2 - z_{253}^2 \\ &- z_{254} - z_{254}^2 \end{aligned}$$

$$R_{185} = -1 + z_{129}^2 - 2\, z_{131}^2 - 2\, z_{135}^2 - z_{255}^2 - z_{256}^2 - z_{257} - z_{257}^2$$

$$R_{186} = -1 + z_{137}^2 - 2\, z_{139}^2 - 2\, z_{143}^2 - z_{258}^2 - z_{259}^2 - z_{260} - z_{260}^2$$